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Answer the following questions:

A. For the continuous casting answer the following:

The Typical products are:

The Material of the mold used is:

The types of continuous casting machines are:

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.....
.....

B. Show what is meant by green sand, dry sand molding sand, backing sand, facing sand, parting sand, and core sand.

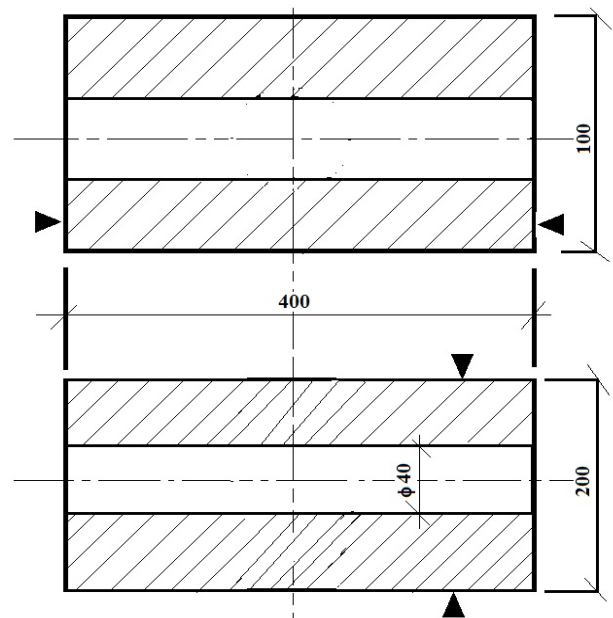
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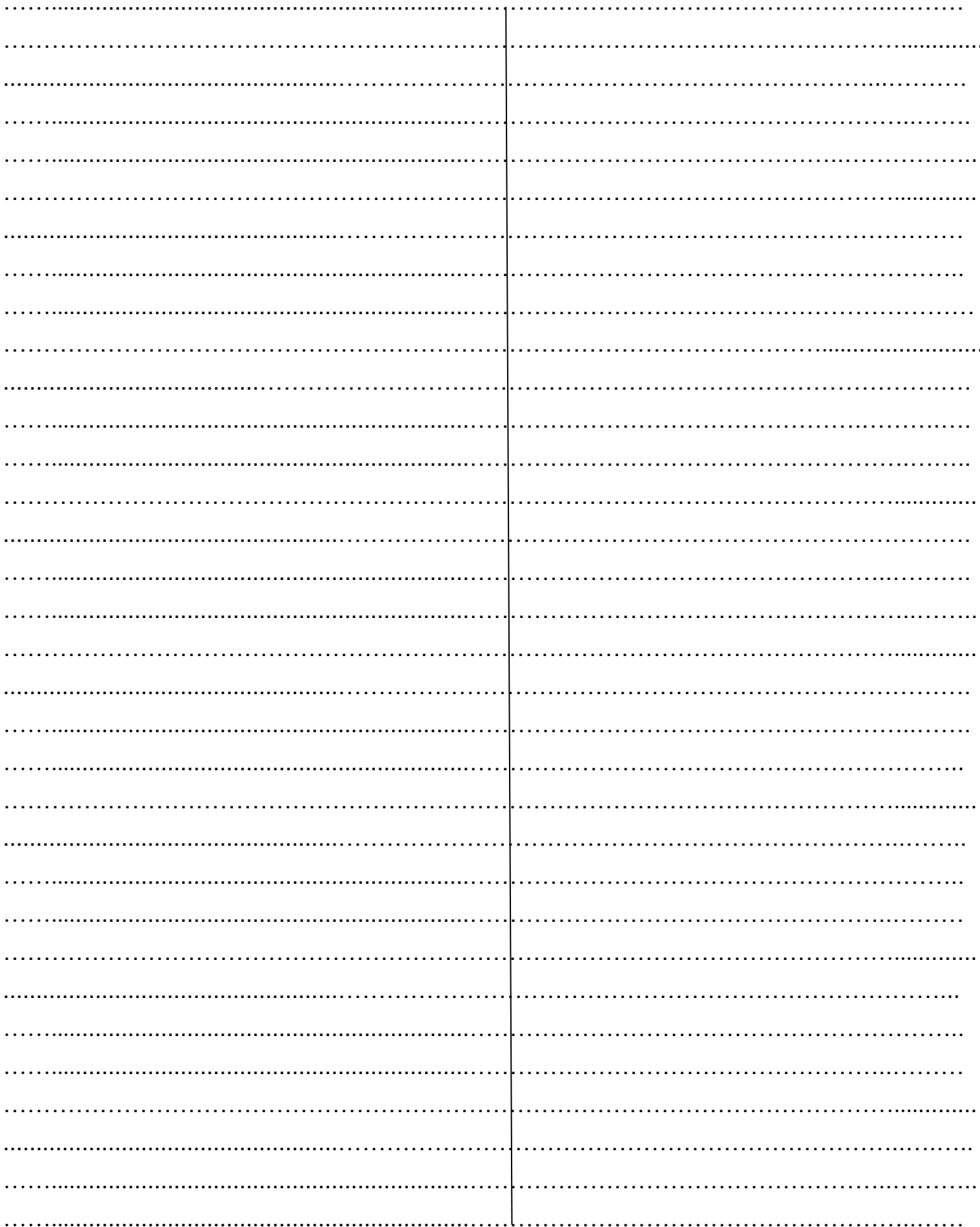
C. It is required to sand cast the shown product from gray cast iron ($\rho = 7.2 \text{ g/cm}^3$), knowing that the product will be machined after casting on the surfaces with (▼) mark only. Find:

1. Suggest suitable casting modifications,
2. Suitable flask dimensions.
3. Core dimensions.
4. Casting pattern dimensions.
5. The forces acting on core.
6. The forces acting on cope.
7. The forces acting on mold.

Take: Shrinkage allowance = 5 mm/m

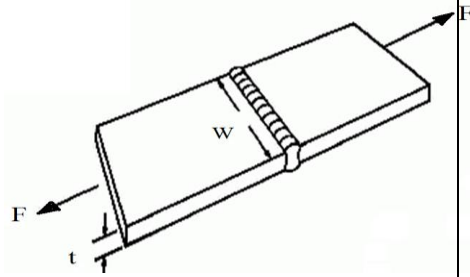
Machining allowance = 3 mm





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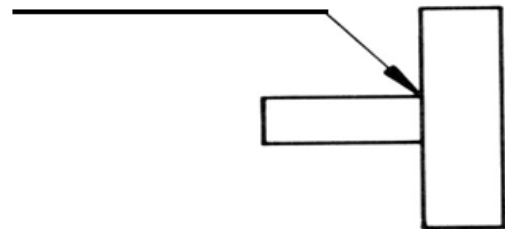
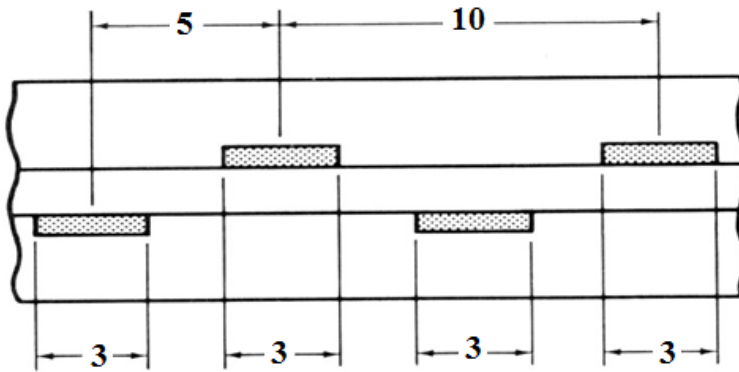
Calculate [F] the force needed to break the Joint, Two steel plates,
[t = 6 mm & W = 30 mm]
SMAW Electrode **E6010**
Note that:
[1 Ksi = 6.89 MPa]
($\tau = 0.6 \sigma$)



2 Marks

Assign the Weld Symbol

2 Marks



Weld size 1/4 in

Illustrate with neat sketch SW
Stud Arc Welding Process Principles

2 Marks

Illustrate with neat sketch PAW
Plasma Arc Welding Process Principles

2 Marks

Define The WPS

2 Marks